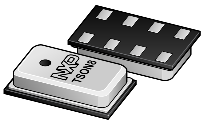

MPL3115A2

I²C precision pressure sensor with altimetry

Rev. 8 — 12 April 2018 Data sheet: technical data

1 General description

The MPL3115A2 is a compact, piezoresistive, absolute pressure sensor with an I²C digital interface. MPL3115A2 has a wide operating range of 20 kPa to 110 kPa, a range that covers all surface elevations on earth. The MEMS is temperature compensated utilizing an on-chip temperature sensor. The pressure and temperature data is fed into a high resolution ADC to provide fully compensated and digitized outputs for pressure in Pascals and temperature in °C. The compensated pressure output can then be converted to altitude, utilizing the formula stated in Section 9.1.3 "Pressure/altitude" provided in meters.The internal processing in MPL3115A2 removes compensation and unit conversion load from the system MCU, simplifying system design.

MPL3115A2's advanced ASIC has multiple user programmable modes such as power saving, interrupt and autonomous data acquisition modes, including programmed acquisition cycle timing, and poll-only modes. Typical active supply current is 40 μA per measurement-second for a stable 10 cm output resolution.

2 Features and benefits

- Operating range: 20 kPa to 110 kPa absolute pressure
- Calibrated range: 50 kPa to 110 kPa absolute pressure
- Calibrated temperature output: −40 °C to 85 °C
- I²C digital output interface
- Fully compensated internally
- Precision ADC resulting in 0.1 meter of effective resolution
- Direct reading
  - Pressure: 20-bit measurement (Pascals)
    - 20 to 110 kPa
  - Altitude: 20-bit measurement (meters)
    - –698 to 11,775 m
  - Temperature: 12-bit measurement (°C)
    - –40 °C to 85 °C
- Programmable interrupts
- Autonomous data acquisition
  - Embedded 32-sample FIFO
  - Data logging up to 12 days using the FIFO
  - One-second to nine-hour data acquisition rate
- 1.95 V to 3.6 V supply voltage, internally regulated
- 1.6 V to 3.6 V digital interface supply voltage
- Operating temperature from −40 °C to +85 °C